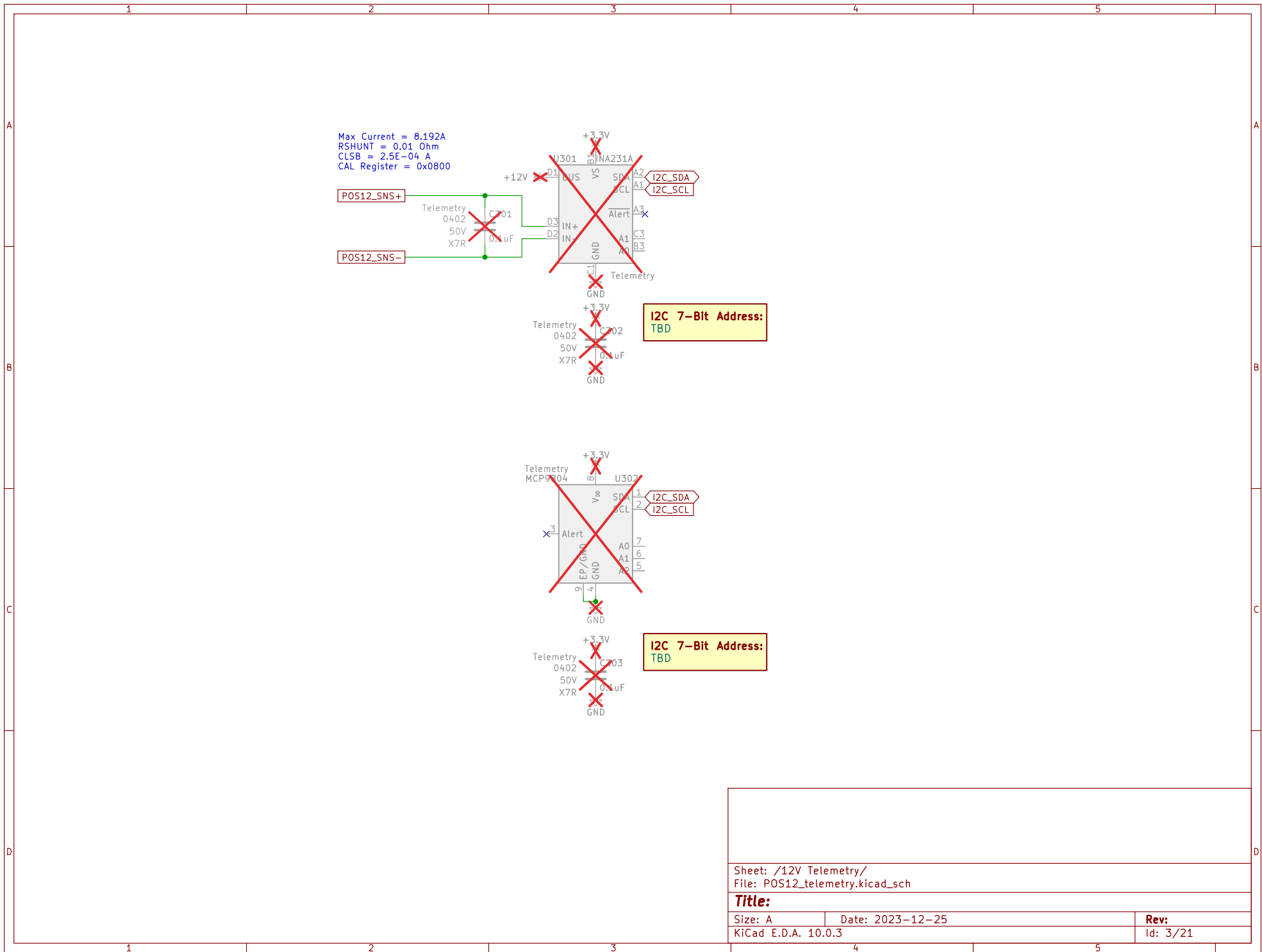
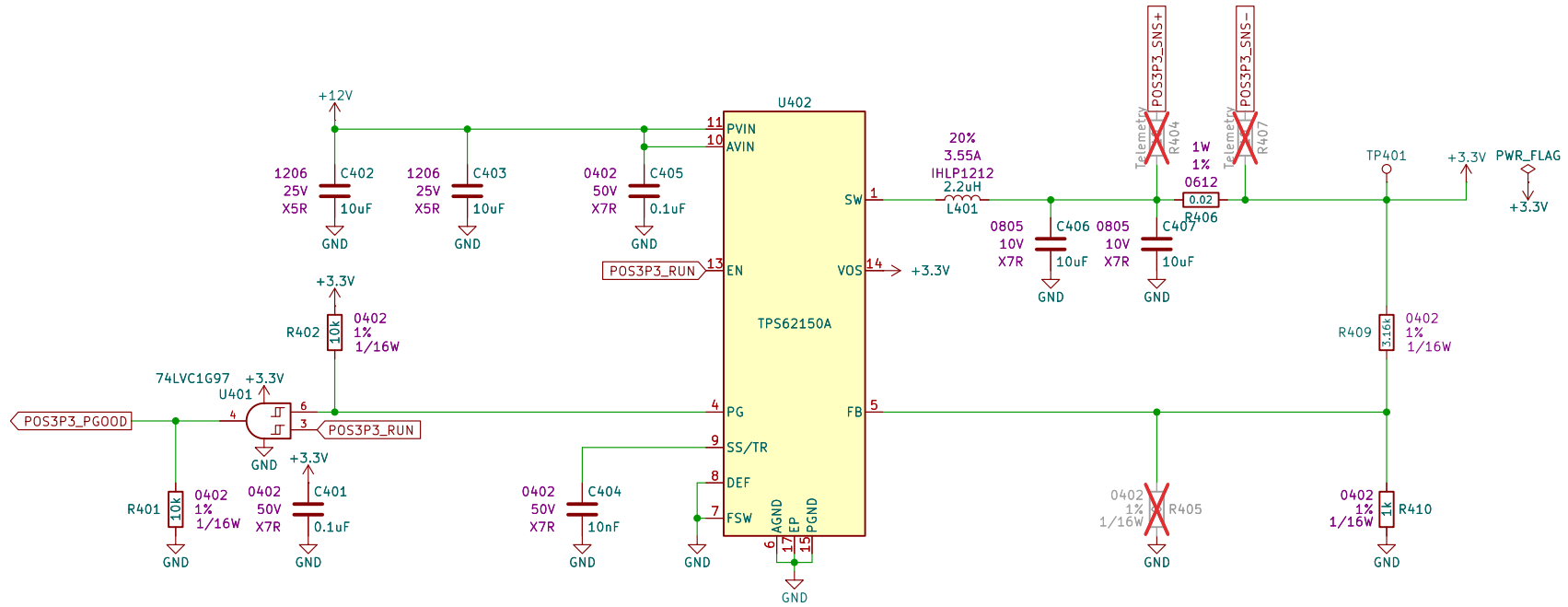


|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |  |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|--|
| 1 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 | 5 |  |
| A | 12V Power Input<br><div></div> File: POS12_power_input.kicad_sch<br>12V Telemetry<br><div></div> File: POS12_telemetry.kicad_sch<br>3.3V Power Supply<br><div></div> File: POS3P3_power_supply.kicad_sch<br>3.3V Telemetry<br><div></div> File: POS3P3_telemetry.kicad_sch<br>dsPIC33 Programming<br><div></div> File: dspic33_programming.kicad_sch<br>dsPIC33 Power<br><div></div> File: dspic33_power.kicad_sch<br>dsPIC33 GPIO<br><div></div> File: dspic33_gpio.kicad_sch<br>USB UART Bridge<br><div></div> File: usb_uart_bridge.kicad_sch | Platform ETC<br><div></div> File: platform_etc.kicad_sch<br>PGOOD LEDs<br><div></div> File: pgood_leds.kicad_sch<br>Status LEDs<br><div></div> File: status_leds.kicad_sch<br>RCA Signal Conditioning<br><div></div> File: rca_signal_conditioning.kicad_sch<br>Speaker Signal Conditioning<br><div></div> File: speaker_signal_conditoning.kicad_sch<br>Source Selection, PGAs<br><div></div> File: source_selection_pgas.kicad_sch<br>Anti-alias Filtering<br><div></div> File: anti_alias_filtering.kicad_sch<br>Audio Voltage References<br><div></div> File: audio_voltage_references.kicad_sch<br>LED Panel Buffers<br><div></div> File: led_panel_buffers.kicad_sch<br>Pushbuttons<br><div></div> File: pushbuttons.kicad_sch<br>Mechanical<br><div></div> File: mechanical.kicad_sch<br>LED Panel Connectors<br><div></div> File: led_panel_connectors.kicad_sch |   |   |  |
| B |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |  |
| C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |  |
| D |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |  |
| 1 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 | 5 |  |

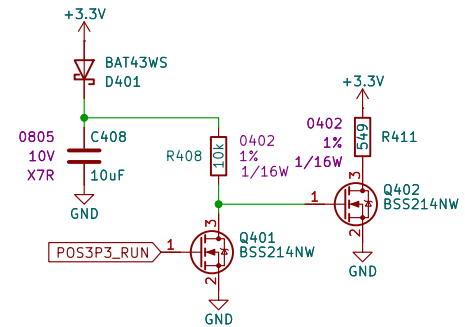






| Load                  | Draw    | Quantity | Total    |
|-----------------------|---------|----------|----------|
| dsPIC                 | 50mA    | 1        | 50mA     |
| 8MHz dsPIC Clock      | 7mA     | 1        | 7mA      |
| MCP6N11               | 1.1mA   | 6        | 6.6mA    |
| MCP6024               | 5.4mA   | 3        | 16.2mA   |
| Analog_Switch         | 0.05mA  | 1        | 0.05mA   |
| Digipot               | 0.6mA   | 1        | 0.6mA    |
| 1.6V_Reference        | 0.2mA   | 1        | 0.2mA    |
| PCA9536_GPIO_Expander | 0.29mA  | 1        | 0.29mA   |
| I2C_Buffers           | 8.3mA   | 3        | 24.9mA   |
| Octal_Buffers         | 1mA     | 2        | 2mA      |
| LVC1G19s              | 0.01mA  | 21       | 0.21mA   |
| Rotary_Encoder_Module | 50mA    | 1        | 50mA     |
| LEDs                  | 1.3mA   | 11       | 14.3mA   |
| INA231                | 0.42mA  | 2        | 0.84mA   |
| MCP9804               | 0.4mA   | 3        | 1.2mA    |
| ETC                   | 0.36mA  | 1        | 0.36mA   |
| Cap_Touch_Sensor      | 0.729mA | 1        | 0.729mA  |
|                       |         |          |          |
|                       |         |          |          |
| Total                 |         | 0.579W   | 175.48mA |

POS12\_PG00D 0 POS3P3\_RUN  
R403

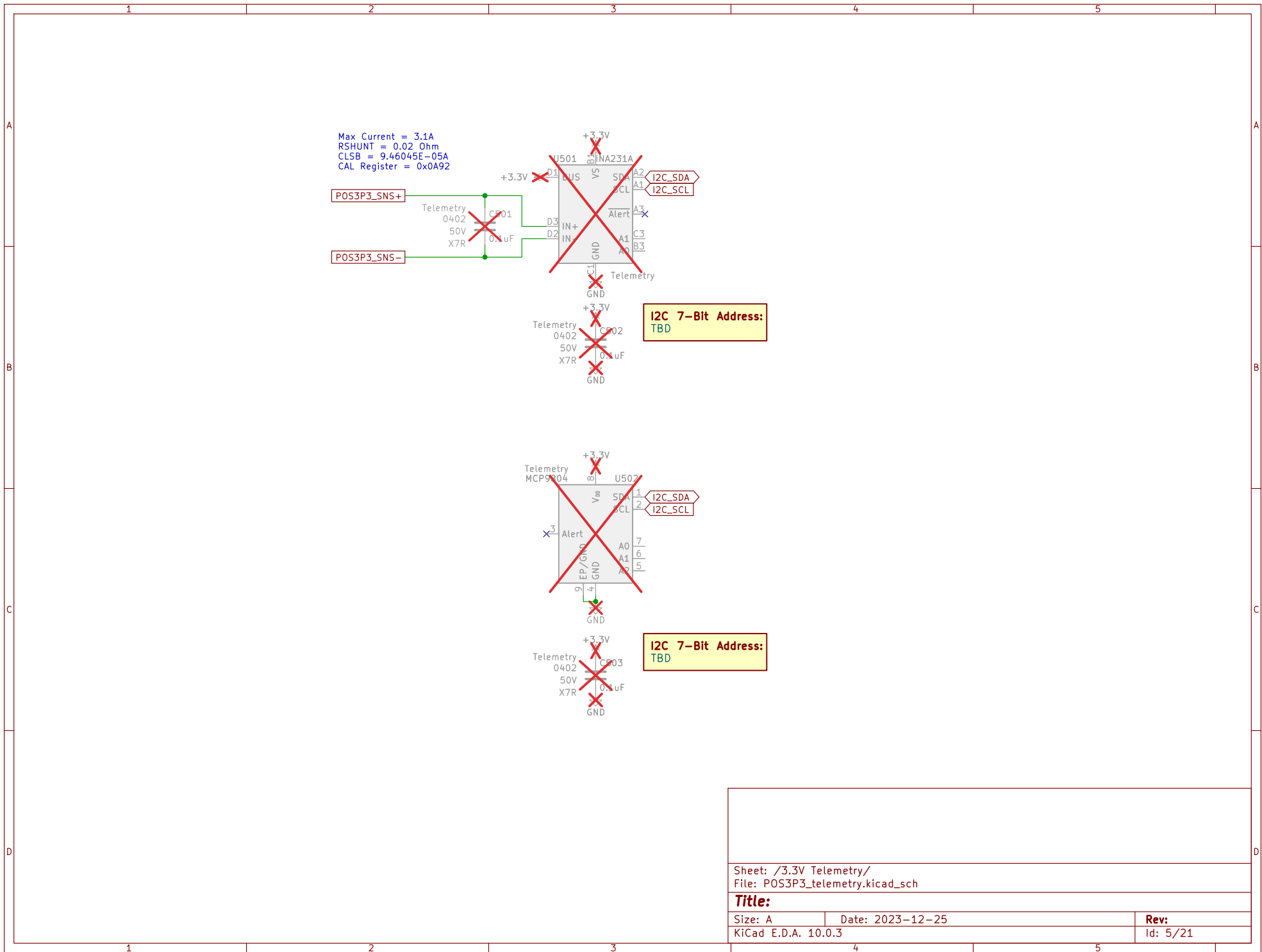


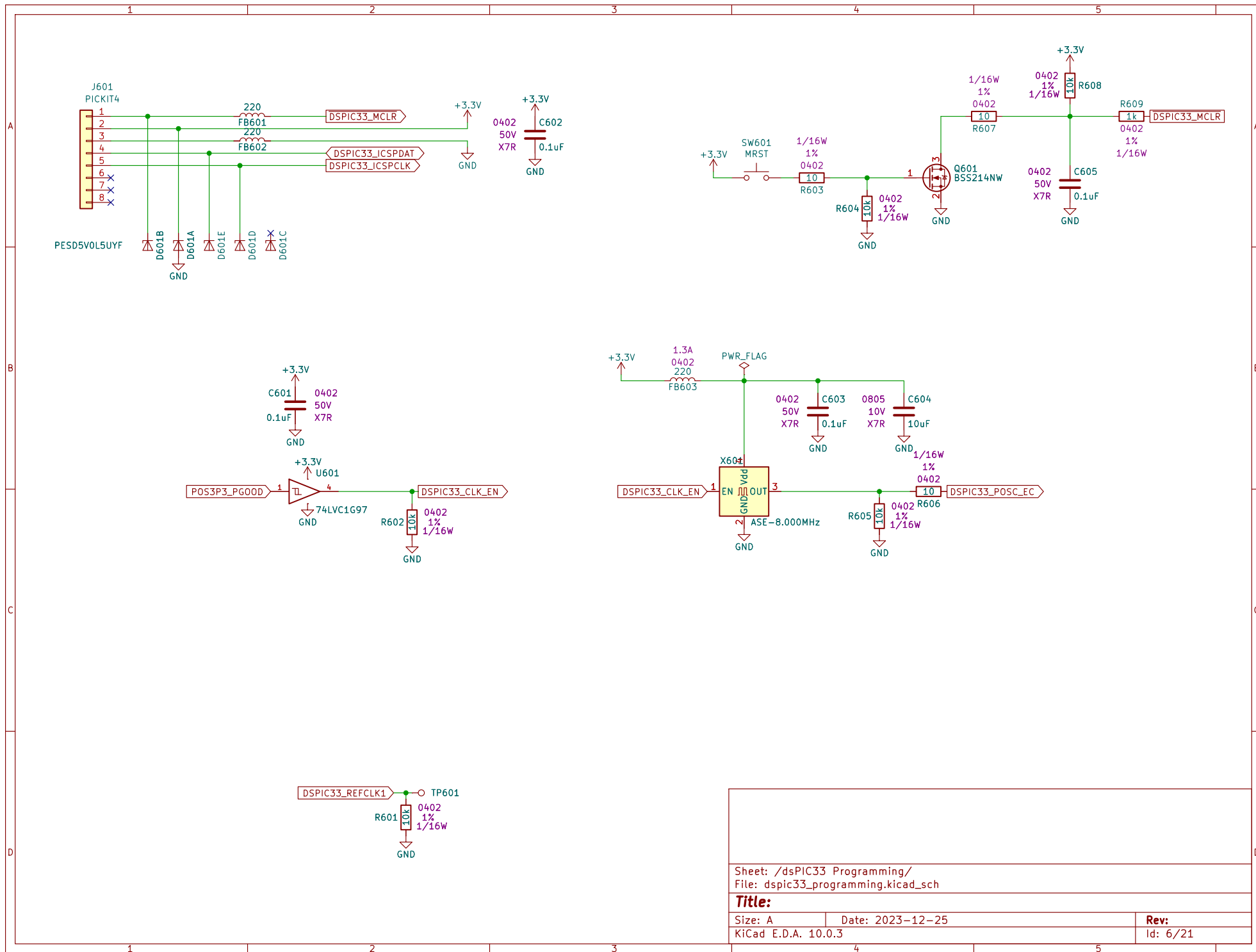
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File: POS3P3\_power\_supply.kicad\_sch

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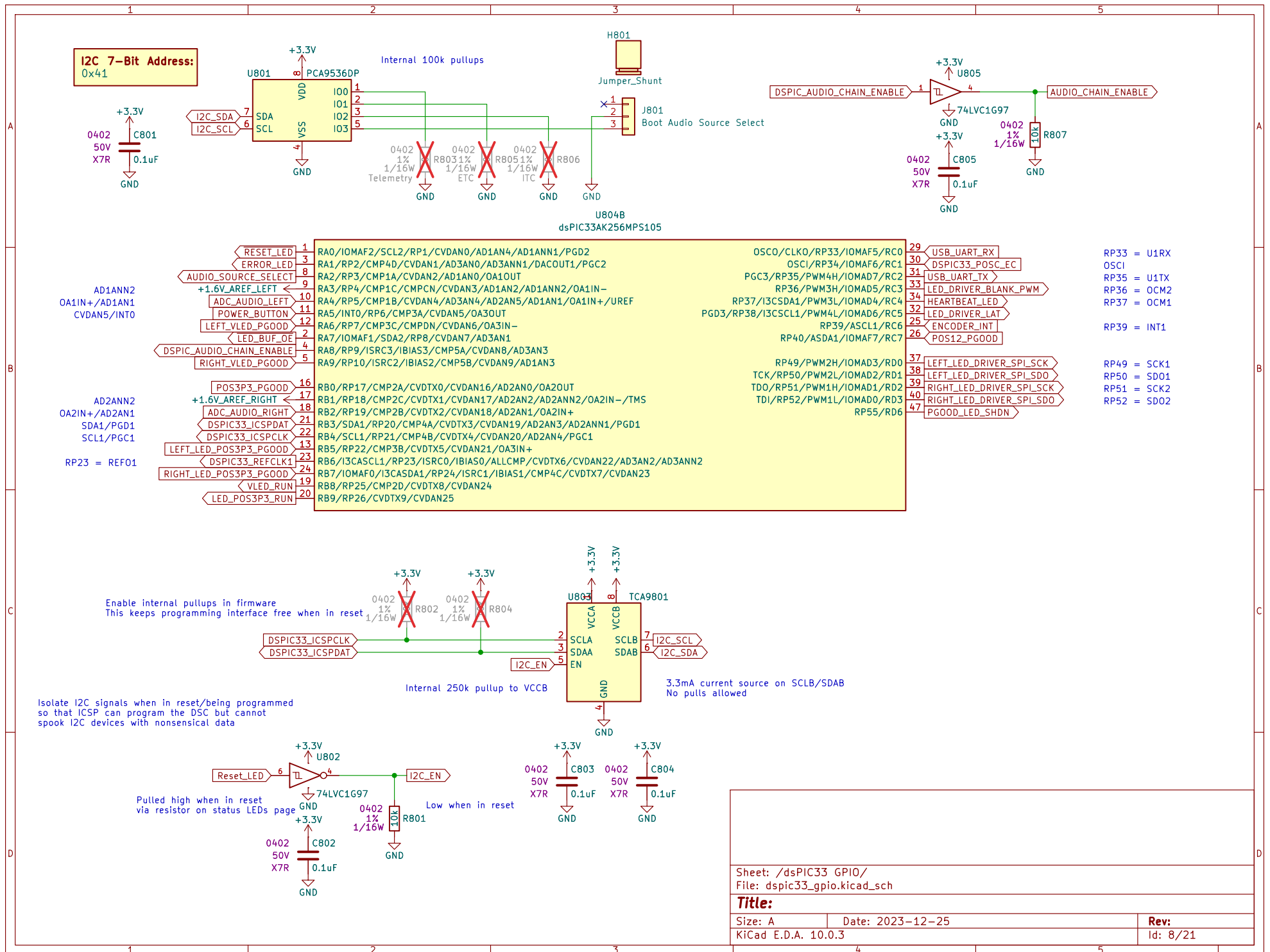
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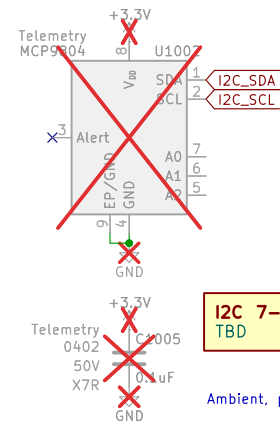
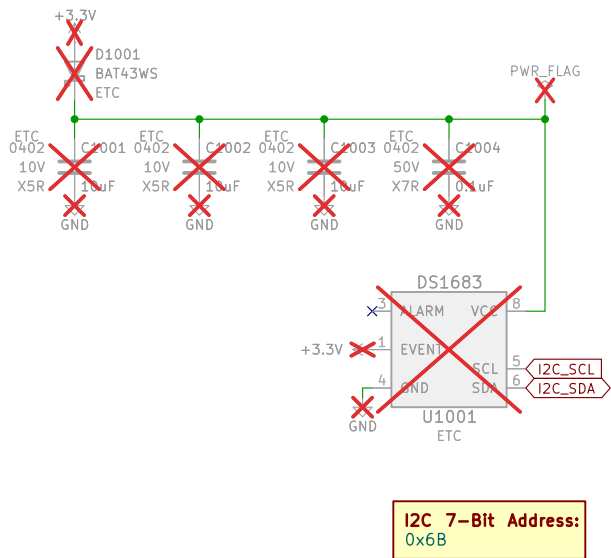




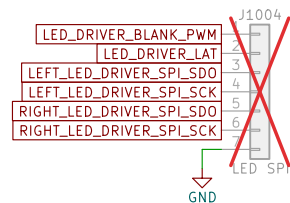
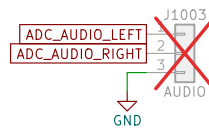
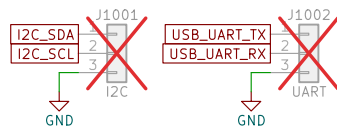








#### Test Points

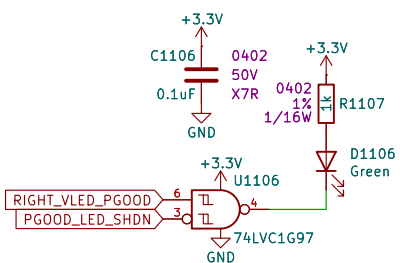
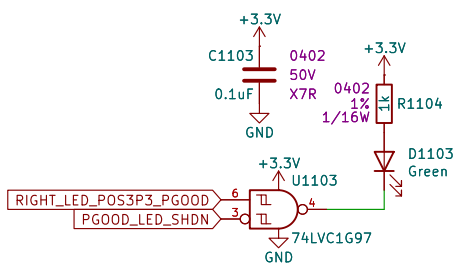
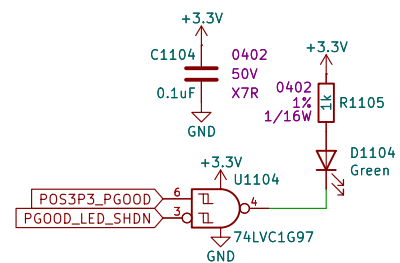
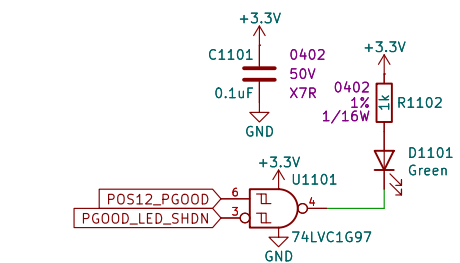


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File: platform\_etc.kicad\_sch

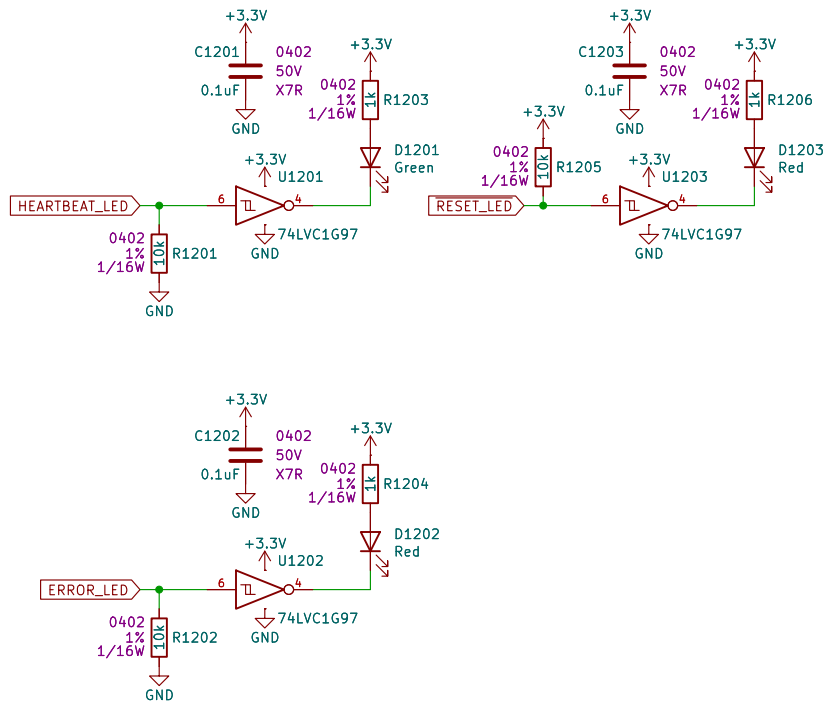
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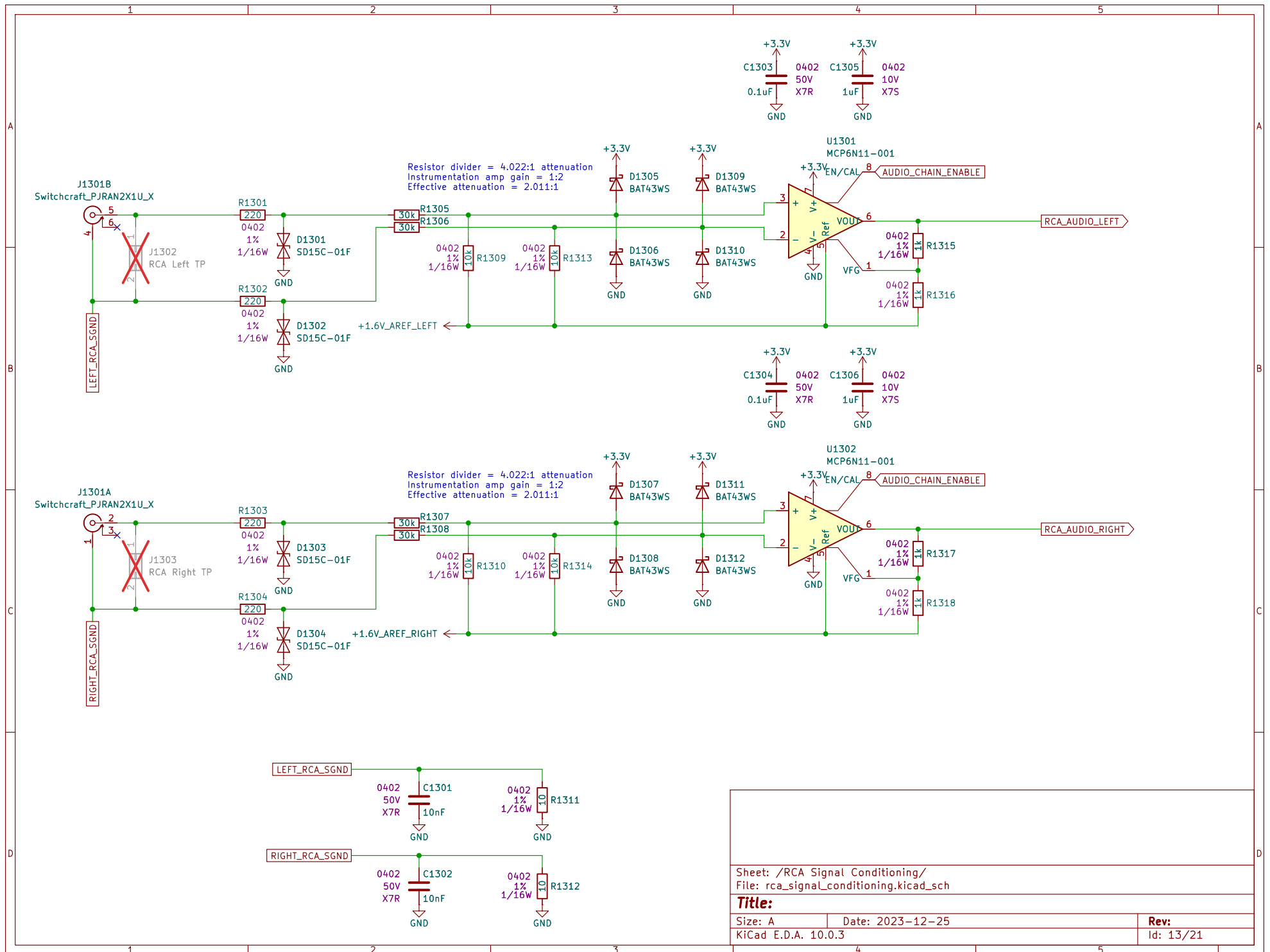


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Size: A Date: 2023-12-25  
KiCad E.D.A. 10.0.3

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Id: 12/21



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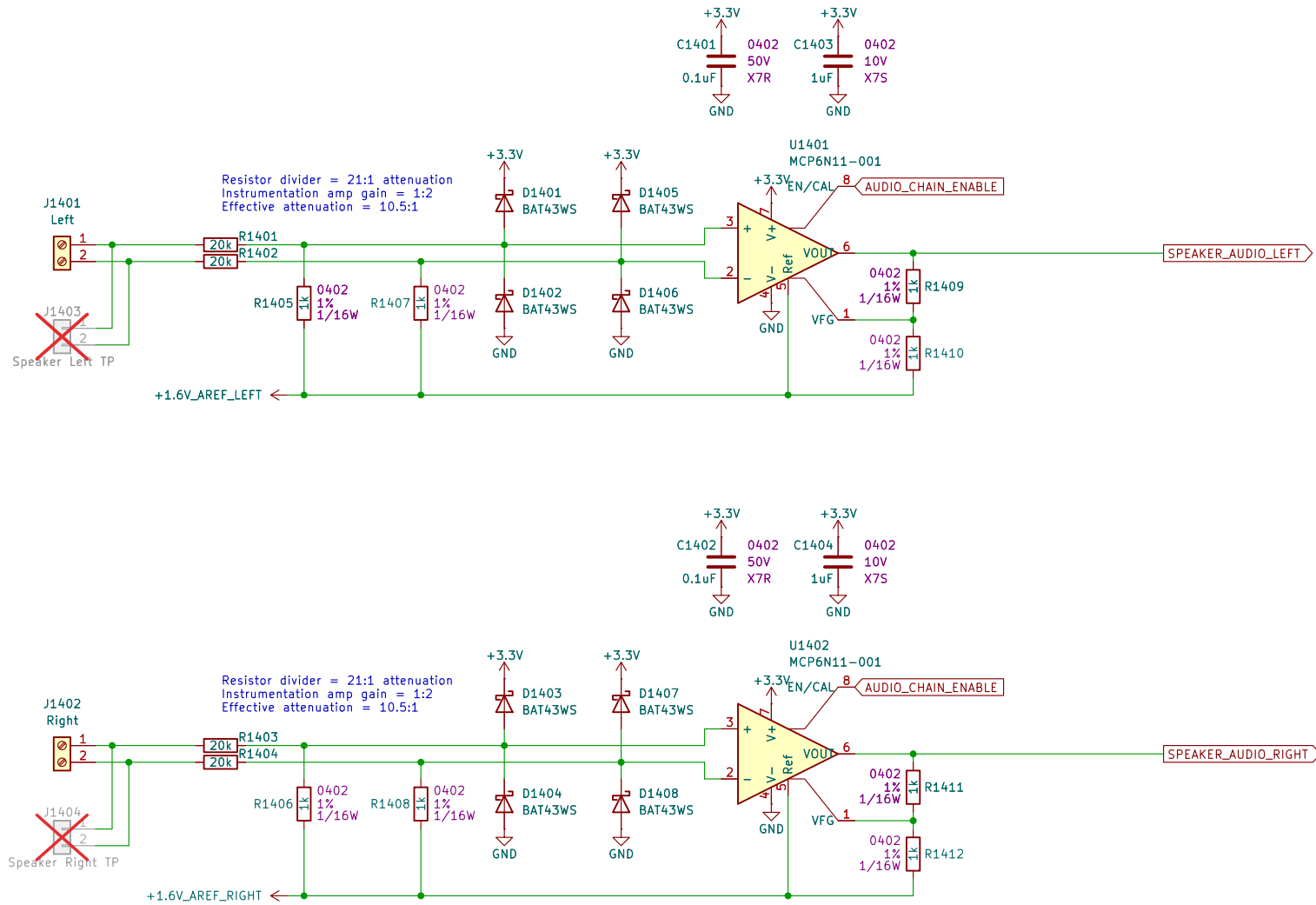
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**Rev:**

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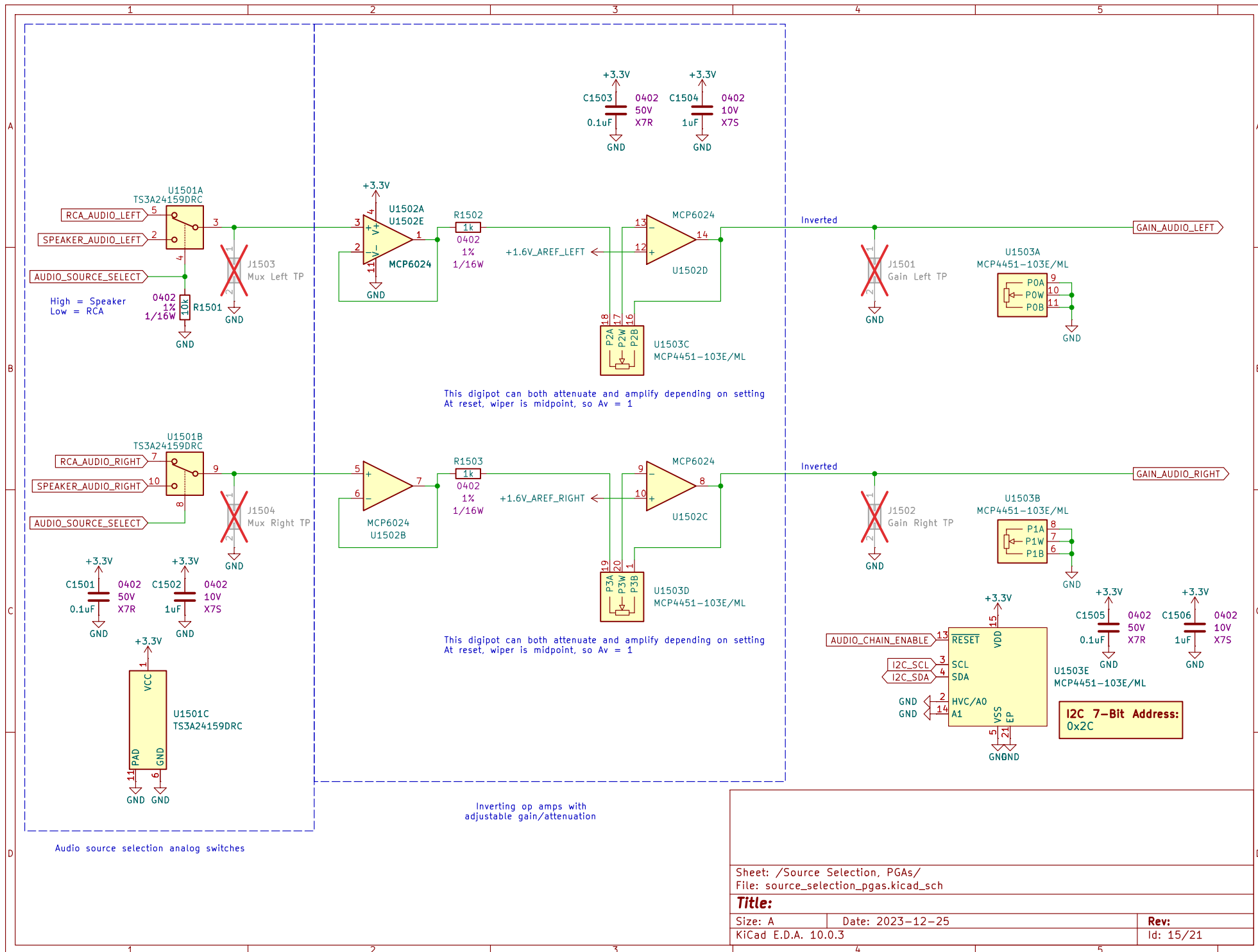
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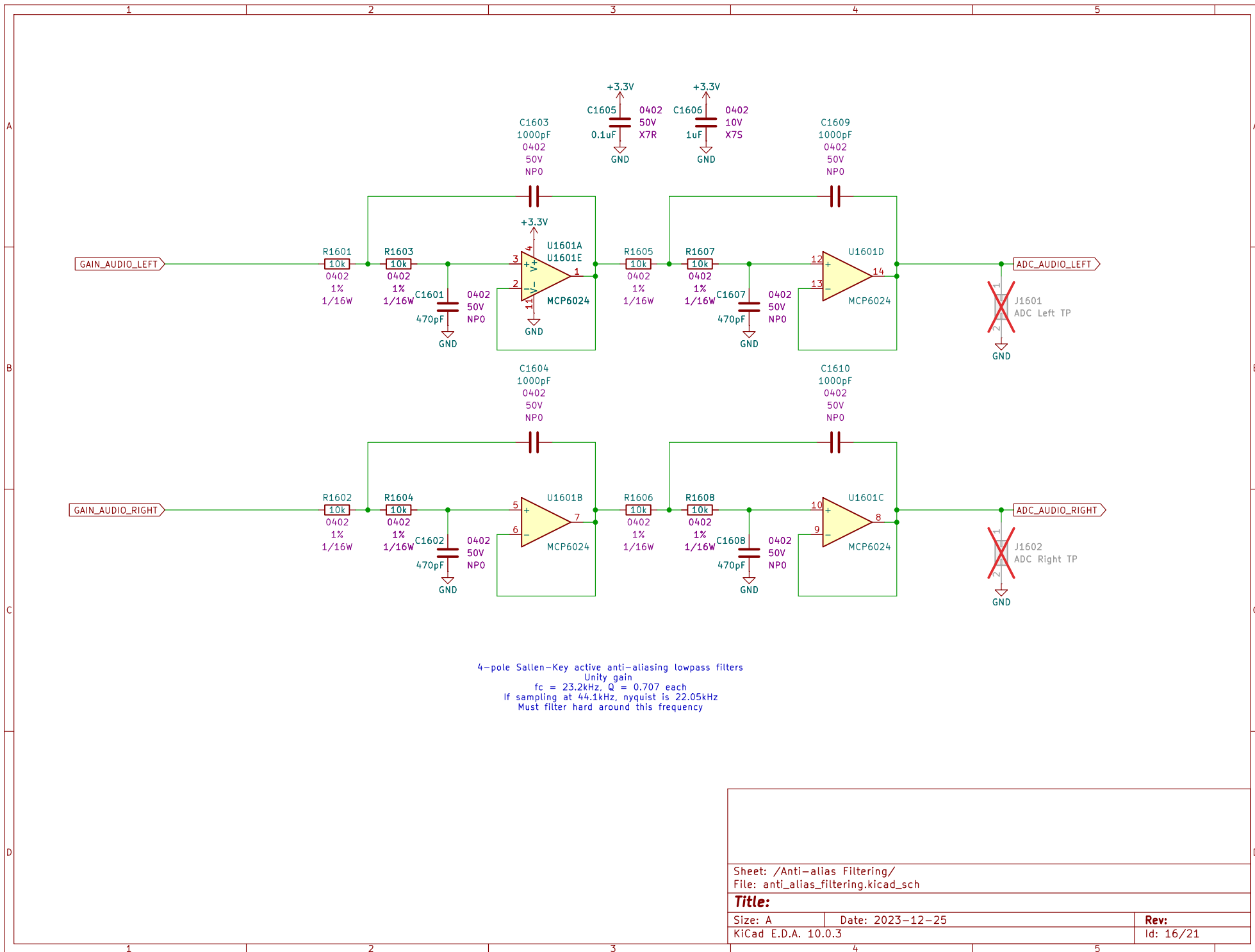
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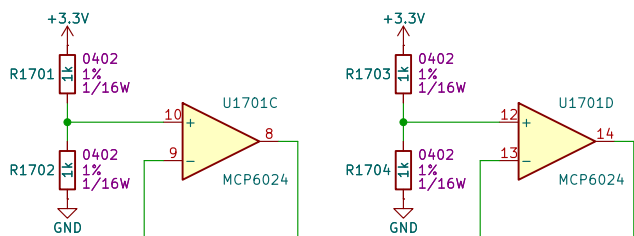
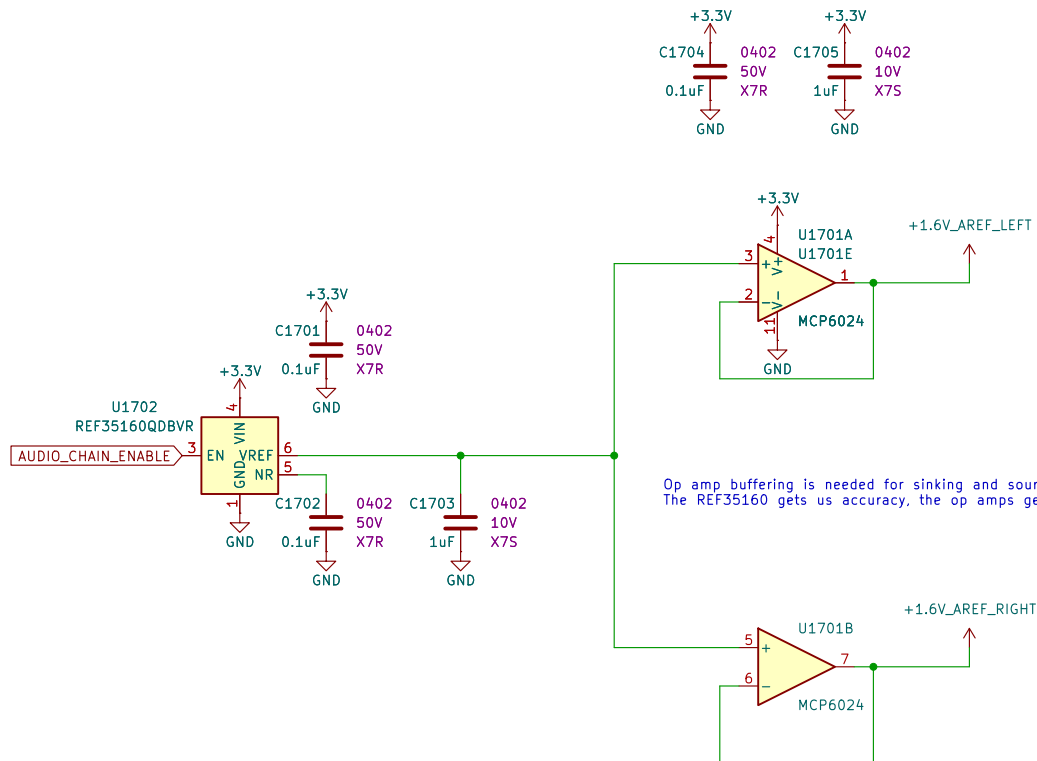
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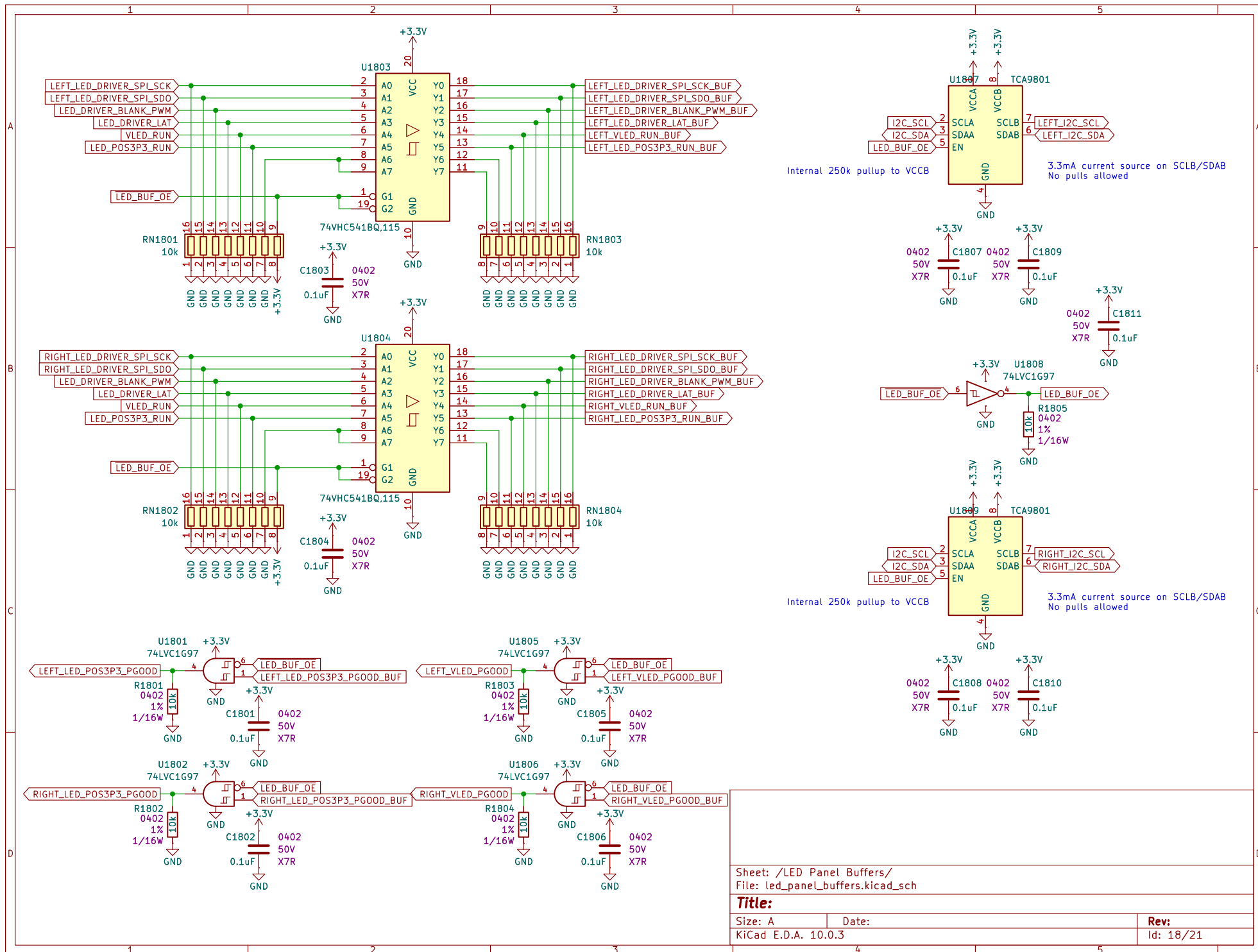


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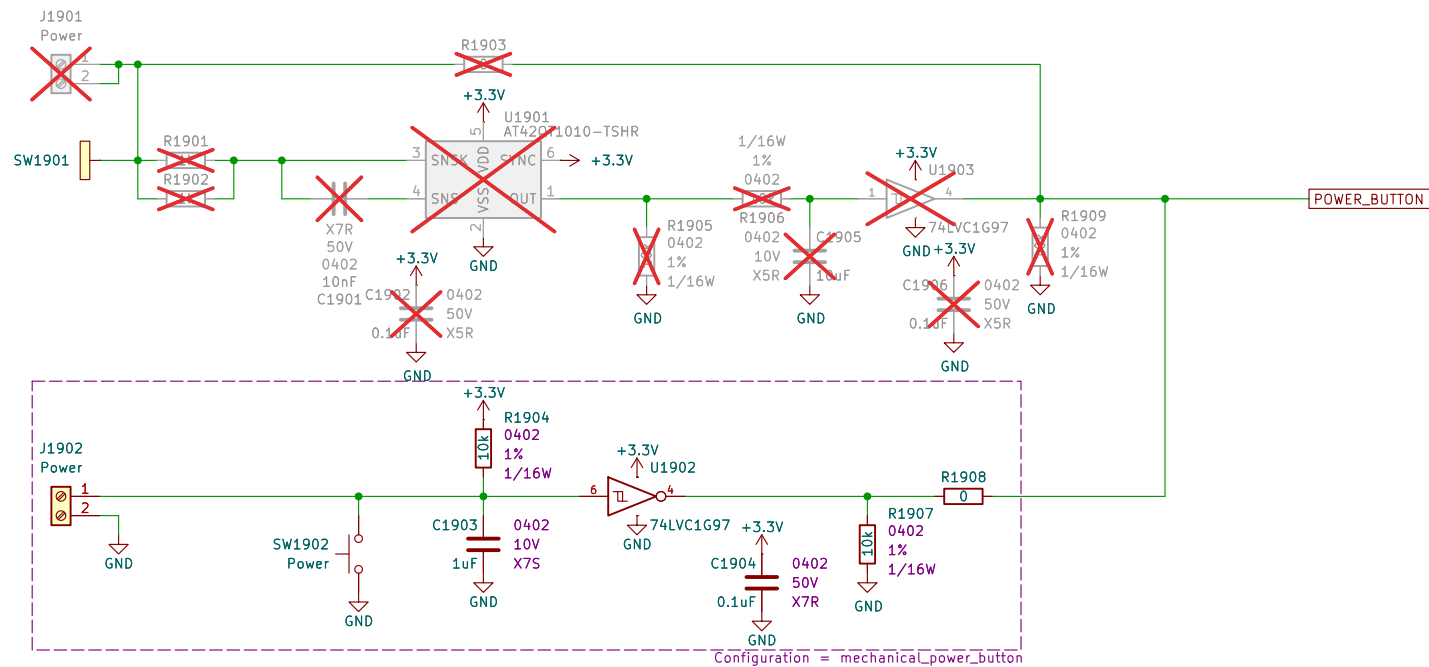
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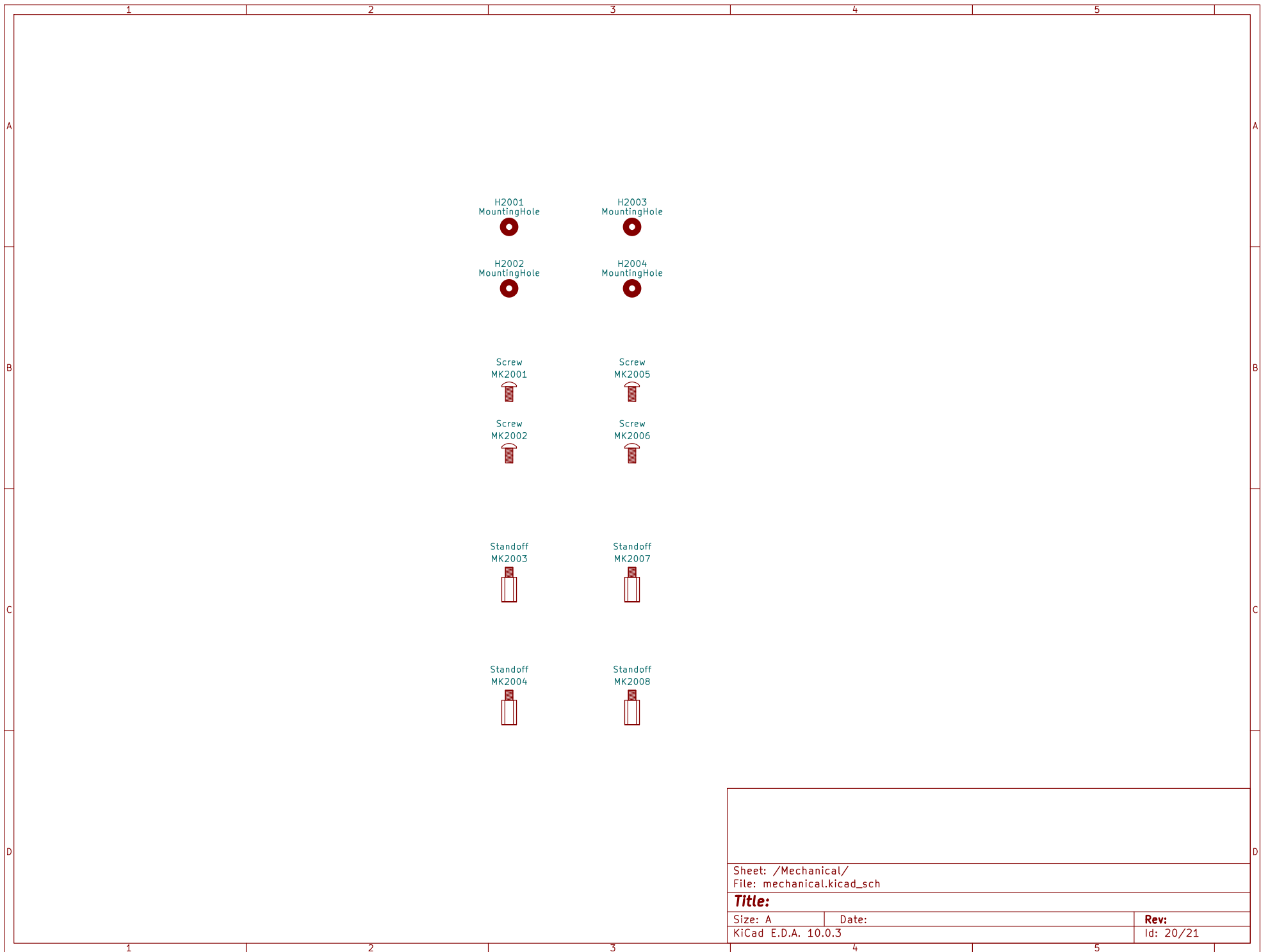


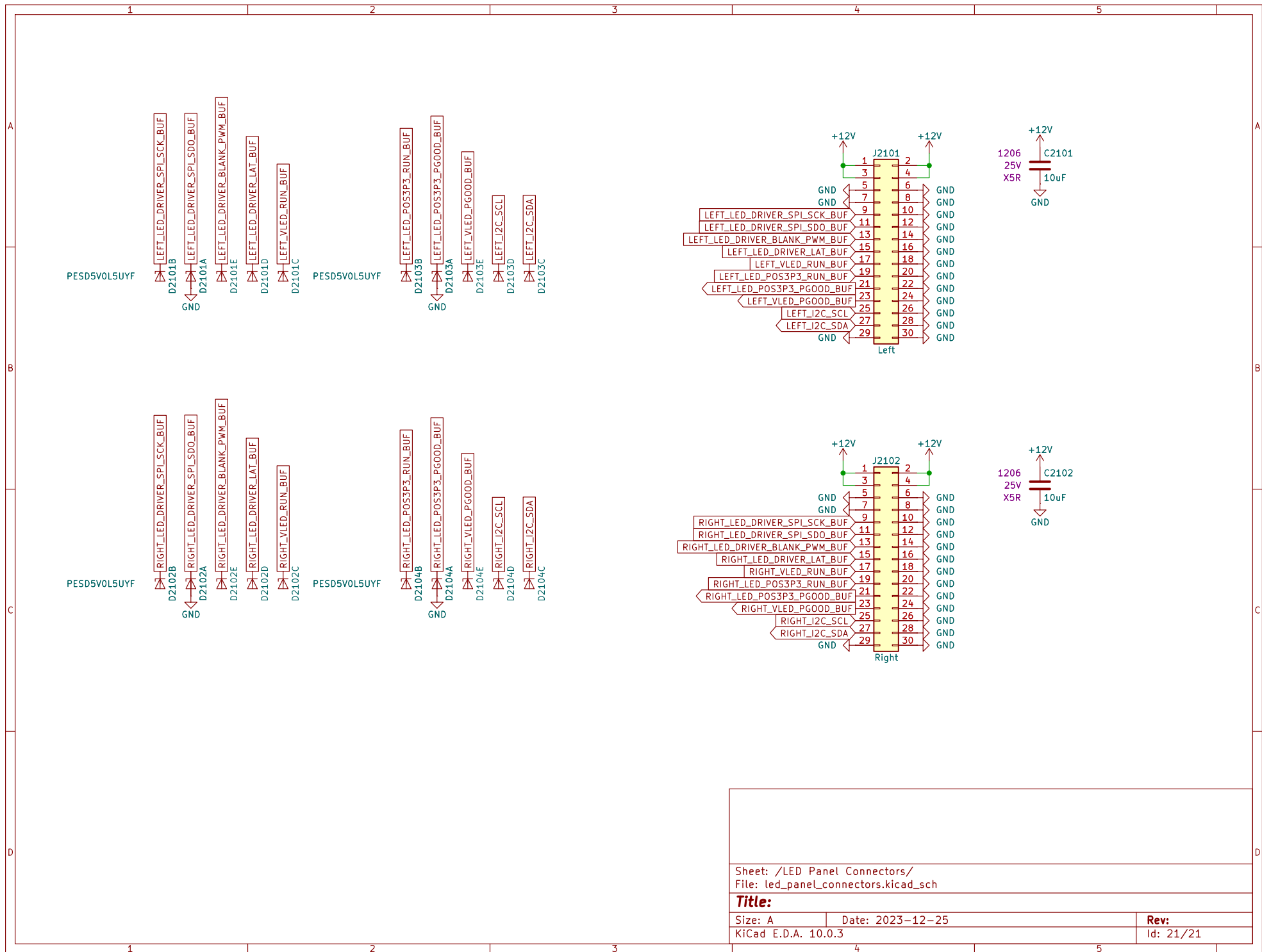
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Size: A Date:  
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Sheet: /LED Panel Connectors/  
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**Title:**

Size: A Date: 2023-12-25  
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Id: 21/21